Mathematical Modelling of Biological Systems

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Outline

- Introduction to mathematical models and why they are useful in biology.
- Simple vs. complex models
- Deterministic vs. stochastic models: advantages and disadvantages of each.

Mathematical modelling

Definition (A mathematical model)

- A simplified representation of a complex systems, which can be used to analyse the system, make predictions, and gain insights into the behaviour of the system.
- In biology, mathematical models help to better understand and analyse biological processes eg. behaviour of cells and dynamics of populations and ecosystems.

Why mathematical models are important.

Importance of mathematical models to biology

- Allow researchers to study and make predictions about complex systems.
- Help identify key factors influencing biological processes.

Simple vs complicated models

Simple models

- Involve a small number of variables and assumptions.
- Easy to develop, understand, and interpret, and usually solve mathematically.
- Do not depict real systems.
- Can sometimes help to gain basic understanding of the system, but its usefulness is limited if it does not describe the real system.

Example: Birth-death model $\left(\frac{dI}{dt} = b - dI\right)$

$$\emptyset \xrightarrow{b} I \xrightarrow{d} \emptyset$$

Simple vs complicated models

Complicated models

- Involve a large number of variables and assumptions.
- Challenging to develop, understand, interpret, and solve mathematically (Sometimes impossible to solve).
- Comprehensive, hence depicts real systems.
- Useful when the goal is to accurately capture the complexity of the system and make detailed predictions about its behaviour.

Example: SIR model $\left(\frac{dS}{dt} = -\beta SI; \frac{dI}{dt} = \beta SI - \gamma I; \frac{dR}{dt} = \gamma I\right)$

How complicated should we go?

Choice on model complexity

Generally depends on the specific research question and the available data and resources.
We however want a model with a good approximation to the reality, and one that we can analyse.

Examples

- Add or split compartments? eg. split infectious into symptomatic and asymptomatic.
- Can a recovered become infected again?
- Should we consider factors such as temperature and location?

We need to strike a balance between simplicity and complexity.

Deterministic vs stochastic models

Deterministic models

- Assumes that the system is completely predictable and there is no randomness or uncertainty involved
- Pros include:
  - Often simple to develop and understand
  - Accurate for large data where randomness tend to average out
  - Provide precise prediction
- Cons include:
  - Inability to account for variability / randomness
  - May not capture the true behaviour of the system, hence limited accuracy.

Deterministic vs stochastic models

Stochastic models

- Takes into account random events and variability in the system, and predicts the probability of certain outcomes.
- Pros include:
  - Ability to account for variability and randomness, small fluctuations, and rare events that may have significant impacts
  - More accurate and real
- Cons include:
  - More complicated hence more difficult to analyse
  - May be computer intensive

Example

Birth-death model

The model:

$$\emptyset \xrightarrow{b} I \xrightarrow{d} \emptyset$$

The deterministic equation:

$$\frac{dI}{dt} = b - dI$$

The stochastic equation:

$$\frac{dP(n,t)}{dt} = -(b + dn)P(n,t) + bP(n-1,t) + d(n+1)P(n+1,t)$$

When to use deterministic or stochastic models

Stochastic or determinitics?

Generally depends on factors such as:

- The specific research question
- Complexity and level of accuracy required
- Available data and computational resources

In general, if the system is large, a deterministic works well since the randomness averages out.
If small, stochastic models are more appropriate since randomness is an important influence .

References

Allen, L. J. (2010). An introduction to stochastic processes with applications to biology. CRC press.

Baker, R. E. (2018). Mathematical modelling in biology: lecture notes. University of Oxford.

THANK YOU-:)